

Project 1: Numerical solutions to 2D parabolic and elliptic PDEs

Present your work in the style of a journal article using scientific languages and support your discussions using appropriate figures. Please use the `aapm` class inside the `REVTeX 4.2e` package as the template of your report. Template can be found on Webcampus.

Important notes before you start:

1. Your report to the project is **limited to 5 pages** using the required template (including references), and longer reports do not translate to better grades. You are required to append your code to your report, and this is not counted towards the 5-page limit. You can simply export/print your code into a single PDF and append it to your report. No need to include that in the L^AT_EX template of your report.
2. Presentation really matters. For scientific papers and formal reports, good presentation is not a subjective matter. It usually comes with guidelines. You are encouraged to ask me for tips and feedbacks to improve the presentation of your document. The more your document looks like an actual scientific paper, the better grade you will get. Your interpretation and presentation of your results are emphasized as much as the results themselves.

Problem Statement

Consider the heat equation with a source term over the two-dimensional domain of $x \in [0, 1]$ and $y \in [0, 1]$,

$$\frac{\partial u}{\partial t} = \alpha \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) + \kappa(x, y), \quad (1)$$

whose steady-state solution $\tilde{u}(x, y)$ satisfies the Poisson equation,

$$\alpha \left(\frac{\partial^2 \tilde{u}}{\partial x^2} + \frac{\partial^2 \tilde{u}}{\partial y^2} \right) = -\kappa(x, y). \quad (2)$$

The thermal diffusivity is $\alpha = 0.005$ and the time-invariant source term is given as

$$\kappa(x, y) = 0.02e^{-[(x-0.7)^2/0.09 + (y-0.6)^2/0.25]}.$$

Both PDEs are subject to the same set of boundary conditions

$$\begin{cases} u = 0.5 - 0.5 \cos(2\pi y) & \text{at } x = 0, \\ \partial_x u = 0 & \text{at } x = 1, \\ \partial_y u = -0.3 & \text{at } y = 0, \\ u = 0.5 + 0.5 \sin(4\pi x - 0.5\pi) & \text{at } y = 1. \end{cases}$$

The steady state solution $\tilde{u}(x, y)$ is shown in figure (1).

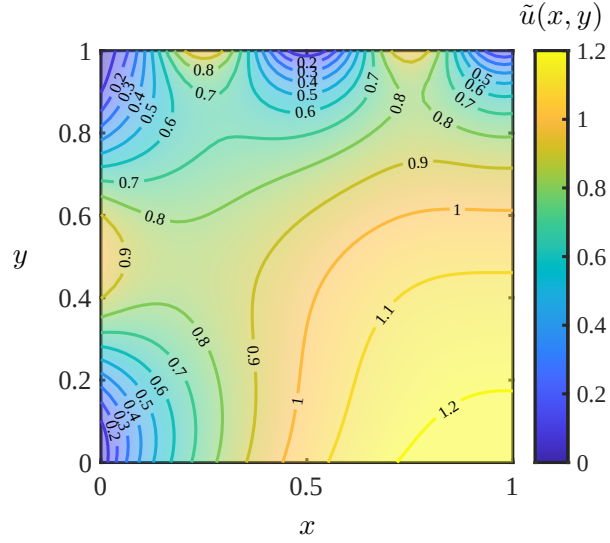


Figure 1: : Steady-state solution $\tilde{u}(x, y)$ obtained from the heat equation (1) and the Poisson equation (2).

Poisson equation

Using 2nd-order central finite differencing and the conjugate gradient method, numerically solve the Poisson equation (2) for the steady state solution $\tilde{u}(x, y)$.

Heat equation for the transient

Using 2nd-order central finite differencing and any time stepping method, solve the heat equation (1) for the transient solution $u(x, y, t)$ marching from the initial condition $u(x, y, 0) = 0$ till

$$\frac{\|u(x, y, t) - \tilde{u}(x, y)\|_2}{\|\tilde{u}(x, y)\|_2} < 10^{-4}.$$

Report the time t at which the solution settles to the steady state according to the criterion above.

Note: No requirements are placed on the mesh grid you use, but 20 to 30 points in each direction should be sufficient to solve this problem.